

Elementary Linear Algebra Course Summary

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1 Key Terms

- **\textbf{System of Linear Equations}**: A set of linear equations with the same variables. The solution to the system is the set of values for the variables that satisfy all equations simultaneously.
- **\textbf{Elimination and Substitution Method}**: A technique for solving systems of linear equations by manipulating the equations to eliminate variables and then substituting the resulting values to find the solution.
- **\textbf{Intersection of Lines/Planes}**: The point(s) where two or more lines or planes intersect. In the context of linear systems, this represents the solution to the system.
- **\textbf{Linear Combination of Vectors}**: A sum of vectors multiplied by scalar values. In the context of linear systems, the solution vector can be expressed as a linear combination of the coefficient vectors.
- **\textbf{Linear Transformation}**: A function that maps vectors from one vector space to another, preserving vector addition and scalar multiplication. Solving a linear system can be viewed as finding the pre-image of a vector under a linear transformation.
- **\textbf{Pre-image}**: The input vector that maps to a given output vector under a linear transformation.
- **\textbf{Image}**: The output vector resulting from applying a linear transformation to an input vector.
- **Coefficient Matrix**: A matrix whose entries are the coefficients of the variables in a system of linear equations.
- **Augmented Matrix**: A matrix formed by appending the column vector of constants to the coefficient matrix of a system of linear equations.
- **Elementary Row Operations**: Operations performed on the rows of a matrix that do not change the solution set of the corresponding system of linear equations. These include swapping two rows, multiplying a row by a non-zero scalar, and adding a multiple of one row to another.
- **Row Echelon Form (REF)**: A matrix in which the leading entry (the first non-zero element) of each row is 1, and all elements below the leading entry in each column are 0.
- **Backward Substitution**: A method for solving a system of linear equations represented by an upper triangular matrix, starting from the last equation and substituting the solutions back into the preceding equations.
- **Pivot Variables**: In a system of linear equations in row echelon form, the variables corresponding to the columns containing the leading 1s.
- **Free Variables**: In a system of linear equations in row echelon form, the variables corresponding to the columns that do not contain a leading 1.

- **Reduced Row Echelon Form (RREF):** A matrix in which the leading entry of each row is 1, all elements below and above the leading entry in each column are 0.
- **Inconsistent System:** A system of linear equations with no solution.
- **$m \times n$ linear System:** A system of m linear equations in n unknowns.
- **Column Space ($\text{Col}(A)$):** The set of all possible linear combinations of the columns of matrix A .
- **Null Space ($\text{Null}(A)$):** The set of all vectors x such that $Ax = 0$.
- **Slack Variables:** Variables added to convert inequalities in a linear programming problem into equalities.
- **Simplex Method:** An iterative algorithm used to solve linear programming problems by moving from one vertex of the feasible region to another, improving the objective function value at each step.
- **Entering Variable:** In the simplex method, the variable with the most negative coefficient in the objective row.
- **Leaving Variable:** In the simplex method, the variable that becomes zero first as the entering variable increases.
- **Pivot Element:** The element at the intersection of the leaving row and entering column in the simplex tableau.
- **Pivot Operation:** The process of updating the simplex tableau by performing row operations to make the pivot element 1 and all other elements in the entering column 0.
- **Feasible Region:** The set of all points satisfying the constraints of a linear programming problem.
- **Homogeneous System:** A linear system of the form $Ax = 0$.
- **Inhomogeneous System:** A linear system of the form $Ax = b$, where b is a non-zero vector.
- **Particular Solution:** A specific solution x_p to the linear system $Ax = b$ such that $Ax_p = b$.
- **$\mathbb{R}^n$:** The set of all n -tuples of real numbers.
- **Vector Addition:** The operation of adding two vectors in $\mathbb{R}^n$ by adding their corresponding components.
- **Scalar Multiplication:** The operation of multiplying a vector in $\mathbb{R}^n$ by a scalar, where each component of the vector is multiplied by the scalar.
- **Zero Vector ($\vec{0}$):** A vector with all components equal to zero.
- **Negative of a Vector ($-\vec{u}$):** A vector obtained by negating each component of a vector $\vec{u}$.
- **Linear Combination:** A vector $\vec{v}$ expressed as a sum of scalar multiples of other vectors $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_k$, i.e., $\vec{v} = c_1\vec{u}_1 + c_2\vec{u}_2 + \dots + c_k\vec{u}_k$, where c_i are scalars.
- **Span:** The set of all possible linear combinations of a given set of vectors.
- **Vector Space:** A set V on which vector addition and scalar multiplication are defined, satisfying eight specific properties (Slides 8, 9).
- **$\mathcal{P}_n$:** The vector space of polynomials of degree at most n .
- **$M_{m \times n}$:** The vector space of all $m \times n$ matrices.
- **Matrix Multiplication:** The operation of multiplying a matrix by a vector or another matrix, resulting in a new vector or matrix. The result is obtained by taking the dot product of each row of the first matrix with the column vector (or each column of the second matrix).

- **Linearity Properties of Matrix Multiplication:** Properties that describe how matrix multiplication interacts with scalar multiplication and vector addition. Specifically, $A(\alpha X + \beta Y) = \alpha AX + \beta AY$ and $(\alpha A + \beta B)X = \alpha AX + \beta BX$, where A and B are matrices, X and Y are vectors, and α and β are scalars.
- **Composition of Linear Transformations:** The sequential application of two or more linear transformations, which can be represented by the multiplication of their corresponding matrices. The composition of transformations A and B is represented by the matrix product AB .
- **Properties of Matrix Multiplication:** A set of rules governing matrix multiplication, including distributivity over addition, the transpose rule, and the non-commutative nature of matrix multiplication ($AB \neq BA$ in general).
- **Identity Matrix:** A square matrix with ones along the main diagonal and zeros elsewhere. Multiplying any matrix by the identity matrix results in the original matrix.
- **Diagonal Matrix:** A square matrix where all elements outside the main diagonal are zero. The product of two diagonal matrices is another diagonal matrix where each diagonal element is the product of the corresponding diagonal elements of the original matrices.
- **Eigenvalues:** The scalar values λ that satisfy the equation $Ax = \lambda x$, where A is a square matrix and x is a non-zero vector.
- **Eigenvectors:** The non-zero vectors x that satisfy the equation $Ax = \lambda x$, where A is a square matrix and λ is an eigenvalue.
- **Diagonalizable Matrix:** A square matrix A that can be expressed in the form $A = QDQ^{-1}$, where Q is an invertible matrix whose columns are the eigenvectors of A , and D is a diagonal matrix whose diagonal entries are the corresponding eigenvalues of A .
- **Change of Basis:** A transformation of a vector from one coordinate system (basis) to another.
- **Markov Chain:** A stochastic model describing a sequence of possible events where the probability of each event depends only on the state attained in the previous event.
- **Transition Matrix:** A square matrix used in Markov chains, where each entry a_{ij} represents the probability of transitioning from state i to state j .
- **Steady-State Distribution:** The long-term probability distribution of a Markov chain, which is independent of the initial state.

2 Problem Types

- **Solving Systems of Linear Equations:** This problem involves finding the values of variables that satisfy a given set of linear equations. Methods like elimination and substitution are used to find the solution, which can be interpreted geometrically as the intersection point of lines (in 2D) or planes (in 3D).
- **Expressing a Vector as a Linear Combination of Other Vectors:** This problem focuses on representing a given vector as a sum of scalar multiples of other vectors. This is closely related to solving systems of linear equations, where the solution vector can be expressed as a linear combination of the column vectors of the coefficient matrix.
- **Understanding Linear Transformations:** This problem involves analyzing a linear transformation that maps vectors from one space to another. Solving a system of linear equations can be viewed as finding the pre-image of a given vector under a specific linear transformation.
- **Determining the solution type of a system of linear equations:** This problem focuses on identifying whether a system of linear equations has a unique solution, infinitely many solutions, or no solution, based on the row echelon form or reduced row echelon form of its augmented matrix.

- **Expressing pivot variables in terms of free variables:** This problem involves expressing the pivot variables of a system of linear equations in terms of its free variables after transforming the system's augmented matrix into row echelon form.
- **Transforming a matrix into row echelon form (REF):** This problem describes the process of applying elementary row operations to transform a matrix into row echelon form, a key step in solving systems of linear equations using Gaussian elimination.
- **Transforming a matrix into reduced row echelon form (RREF):** This problem involves using elementary row operations to transform a matrix into reduced row echelon form, which provides a simplified representation of the solution to a system of linear equations.
- **Determining the number of free variables in a system of linear equations:** This problem involves finding the number of free variables in a system of linear equations, which determines the dimension of the solution space. Examples include E_{x3} , E_{x4} , E_{x6} , E_{x7} , E_{x9} , and E_{x10} .
- **Geometric interpretation of solutions to systems of linear equations:** This problem involves visualizing the solution set of a system of linear equations geometrically. For example, a unique solution corresponds to the intersection of lines or planes at a single point, infinitely many solutions correspond to coincident lines or planes intersecting at a line, and no solution corresponds to parallel lines or planes with no intersection. Examples include E_{x1} , E_{x2} , E_{x3} , E_{x4} , E_{x5} , E_{x6} , E_{x7} , E_{x8} , E_{x9} , and E_{x10} .
- **Converting to Standard Form:** The process of transforming a linear programming problem into a standard form, involving the introduction of slack variables to convert inequalities to equalities and ensuring all variables are non-negative.
- **Pivot Selection Rules:** The rules for selecting the entering and leaving variables during a pivot operation in the simplex method, involving identifying the most negative coefficient in the objective row and the smallest non-negative ratio of RHS values to coefficients of the entering variable.
- **Performing the Pivot Operation:** The steps involved in updating the simplex tableau during a pivot operation, including dividing the leaving row by the pivot element and using row operations to make all other elements in the entering column zero.
- **Geometric Interpretation of the Simplex Method:** The visualization of the simplex method as an iterative process of moving from one vertex of the feasible region to another, improving the objective function value at each step.
- **Column Space and the Solution of $Ax = b$:** The relationship between the column space of a matrix A and the solvability of the linear system $Ax = b$, stating that a solution exists if and only if b is a linear combination of the columns of A .
- **Geometric Interpretation of $Ax = b$:** The geometric representation of the linear system $Ax = b$, where b is a linear combination of the columns of A , and the solution x represents the weights of this combination.
- **Feasible Region and Linear Combinations:** The connection between the feasible region in linear programming and linear combinations, where points in the feasible region are linear combinations of the columns of the constraint matrix A , with coefficients being the decision variables.
- **Homogeneous vs. Inhomogeneous Systems:** The distinction between homogeneous ($Ax = 0$) and inhomogeneous ($Ax = b$) linear systems, focusing on their solution spaces and the condition for the existence of solutions in inhomogeneous systems.
- **Finding a Particular Solution:** The process of finding a vector x_p that satisfies $Ax_p = b$, emphasizing the condition that b must be in the column space of A for a solution to exist.
- **Geometric Interpretation of the Solution to $Ax = b$:** The representation of the general solution to $Ax = b$ as the sum of a particular solution and a solution to the homogeneous system $Ax = 0$, describing the solution set as a shifted null space.

- **General Solution to $Ax = b$:** The expression of the general solution to $Ax = b$ as $x = x_p + x_h$, where x_p is a particular solution and x_h is a solution to the homogeneous system $Ax = 0$.
- **Uniqueness and Existence of Solutions:** The conditions for the existence and uniqueness of solutions to $Ax = b$, stating that existence depends on b being in the column space of A and uniqueness depends on the null space of A containing only the zero vector.
- **Determining if a vector is a linear combination of other vectors:** This problem involves determining if a given vector can be expressed as a linear combination of a set of vectors. This is done by solving a system of linear equations.
- **Determining if a vector is in the span of a set of vectors:** This problem asks whether a given vector belongs to the set of all possible linear combinations of a set of vectors.
- **Verifying the properties of a vector space:** This problem involves checking if a given set with defined vector addition and scalar multiplication satisfies the eight properties of a vector space.
- **Determining Linear Dependence/Independence:** Determining whether a set of vectors is linearly dependent or independent, and identifying relationships between vectors if they are dependent.
- **Finding Redundant Vectors and Reducing a Set:** Identifying and removing redundant vectors from a set while preserving the span of the original set.
- **Consequences of Linear Dependence on Vector Representation:** Demonstrating that if a set of vectors is linearly dependent, any vector in its span can be represented as a linear combination of the vectors in infinitely many ways.
- **Determining the dimension of a vector space:** This problem involves finding the number of vectors in a basis for a given vector space. The dimension is invariant across all bases of the same vector space.
- **Determining linear dependence/independence of vectors:** This problem focuses on establishing whether a set of vectors can be expressed as a linear combination of each other, where the coefficients are not all zero.
- **Finding a basis for a vector space:** This problem involves identifying a set of linearly independent vectors that span the entire vector space.
- **Determining the minimum number of vectors to span a vector space:** This problem focuses on finding the smallest number of vectors required to generate all vectors within a given vector space.
- **Determining the maximum number of linearly independent vectors in a vector space:** This problem involves finding the largest set of vectors within a given vector space that are linearly independent.
- **Understanding Matrix Multiplication:** This problem focuses on understanding the mechanics of matrix multiplication, including the computation of individual elements and the overall dimensions of the resulting matrix.
- **Understanding Linear Transformations and their Composition:** This problem involves understanding how matrix multiplication represents linear transformations and how the composition of multiple transformations can be represented by matrix multiplication.
- **Utilizing Properties of Matrix Multiplication:** This problem involves applying the properties of matrix multiplication, such as distributivity, the transpose property, and the non-commutativity property, to simplify calculations or solve problems involving matrices.
- **Diagonalizing a Matrix:** Finding a matrix Q and a diagonal matrix D such that $A = QDQ^{-1}$, where A is the given matrix, Q is a matrix whose columns are the eigenvectors of A , and D is a diagonal matrix whose diagonal entries are the eigenvalues of A .

- **Computing Matrix Powers using Diagonalization:** Efficiently computing A^k for a large integer k by using the diagonalization $A = QDQ^{-1}$ to obtain $A^k = QD^kQ^{-1}$.
- **Analyzing Markov Chains using Diagonalization:** Using diagonalization to find the steady-state distribution of a Markov chain by analyzing the eigenvalues and eigenvectors of the transition matrix.

3 Algorithm Solutions

- **Gaussian Elimination:** This algorithm solves a system of linear equations by performing elementary row operations to transform the augmented matrix into row echelon form, then back-substitution to find the solution.
- **Substitution Method:** This algorithm solves a system of linear equations by solving one equation for one variable in terms of the others, and then substituting this expression into the other equations to reduce the number of variables. This process is repeated until a single equation with one variable remains, which is then solved. The solution is then back-substituted into the previous equations to find the values of the other variables.
- **Linear Combination of Vectors:** This algorithm expresses a given vector as a linear combination of other vectors by solving a system of linear equations whose coefficients are the weights of the linear combination.
- **Linear Transformation:** This algorithm maps a vector from one vector space to another using a matrix multiplication. Solving a system of linear equations can be viewed as finding the pre-image of a given vector under a linear transformation.
- **Simplex Method:** This algorithm iteratively improves a feasible solution to a linear programming problem by moving from one vertex of the feasible region to another, improving the objective function value at each step until an optimal solution is found. The algorithm uses a tableau to represent the problem and pivot operations to move between vertices.
- **Pivot Operation:** This operation updates the simplex tableau by selecting an entering and leaving variable, and then performing row operations to make the pivot element 1 and all other elements in the entering variable's column 0.
- **Pivot Selection Rules:** The entering variable is chosen as the variable with the most negative coefficient in the objective row. The leaving variable is chosen as the variable whose constraint row has the smallest non-negative ratio of the RHS value to the corresponding coefficient of the entering variable.
- **Vector Addition:** $\vec{u} + \vec{v} = (x)_1 + y_1$
 $x_2 + y_2$
 $\vdots$
 $x_n + y_n$
- **Scalar Multiplication:** $\alpha \vec{u} = (\alpha) x_1$
 αx_2
 $\vdots$
 αx_n
- **Determining if $\vec{v}$ is in the Span of $\{\vec{u}_1, \dots, \vec{u}_k\}$:** Find $c_1, \dots, c_k$ such that $\vec{v} = c_1\vec{u}_1 + c_2\vec{u}_2 + \dots + c_k\vec{u}_k$
- **Eliminating Redundant Vectors:** Solve the dependency equation $c_1u_1 + c_2u_2 + \dots + c_ku_k = \vec{0}$ for $c_1, \dots, c_k$. Discard vectors u_i corresponding to free variables c_i . The remaining vectors are linearly independent.

- **Matrix Multiplication:** $(AX)_i = \sum_{k=1}^n a_{ik}x_k$
- **Composition of Linear Transformations:** $A(BX) = (AB)X$
- **Multiplication of Diagonal Matrices:** If A and B are diagonal matrices, then $(AB)_{ii} = a_{ii}b_{ii}$ and $(AB)_{ij} = 0$ for $i \neq j$.
- **Diagonalization of a Matrix:** $A = QDQ^{-1}$, where A is the matrix to be diagonalized, Q is the matrix of eigenvectors, and D is the diagonal matrix of eigenvalues.
- **Computation of A^k :** $A^k = QD^kQ^{-1}$, where A is a diagonalizable matrix, Q is the matrix of eigenvectors, D is the diagonal matrix of eigenvalues, and k is the power.
- **Markov Chain Steady-State Analysis:** Find the eigenvalues and eigenvectors of the transition matrix A . The eigenvector corresponding to the eigenvalue 1 represents the steady-state distribution.